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Education Project Write Up

Abstract

I investigated whether school performance, measured by average ACT score, can be predicted from socioeconomic factors such as median household income, unemployment rate, percent of students receiving free/reduced price lunch, and the student teacher ratio. Using cleaned district/school data and ordinary least squares regression, I found that percent free/reduced lunch is the strongest single predictor with an r-squared of about 0.61 when modeled alone. In multivariate models, a reduced linear model with unemployment rate, percent college, and percent lunch explains about 62% of the variance. Quadratic terms for median income and unemployment are statistically significant but do not improve out of sample interpretability over linear fits by enough to be worth using. The student teacher ratio shows weak association with ACT after cleaning with an r-squared of about 0.01, likely due to measurement and design factors of that specific data.

Introduction

Socioeconomic context strongly relates to academic achievement, but the relative contribution of specific factors is debated. My analysis aims to understand how well do median household income, unemployment rate, percent of students receiving free/reduced-price lunch, and the student–teacher ratios predict average ACT scores? For this project I combined EDGAP data containing average ACT or SAT-equivalent, NCES identifiers and descriptors, and Common Core of Data school characteristics. The analysis focuses on high schools. CCD and EDGAP datasets were left joined by NCESSCH while the NCES data was left joined with zip code and state. Implausible values and outliers were then removed. Remaining numeric gaps were filled via iterative imputation for the variables except student-teacher ratios. Variables were standardized as needed for effect/size comparisons. I then fit linear and quadratic models, used residual diagnostics and nested-model ANOVA, and compare accuracy with r-squared and mean absolute error. The aim was to quantify the strength and form of relationships between these socioeconomic predictors and ACT performance using EDGAP, NCES, and CCD datasets.

Theoretical Background

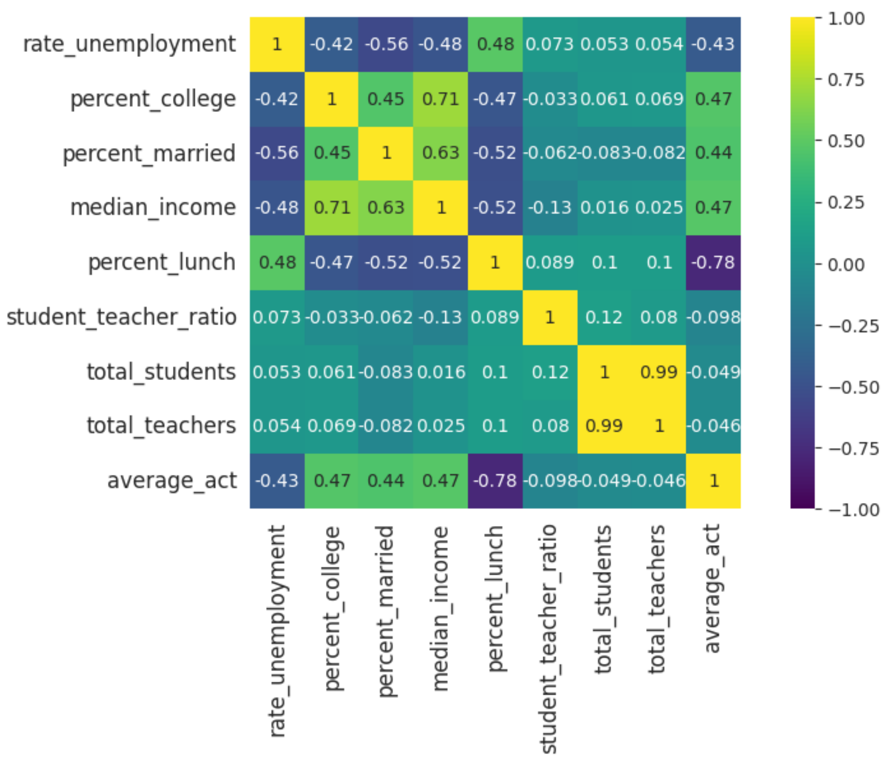
Student achievement is shaped by socioeconomic status through resources, stability, and learning opportunities inside and outside of school. Several of these factors are associated with performance, including median household income and adult college educational. The expected relationship is to be positive with ACT. There is also economic stress with factors such as unemployment rate and free and reduced lunch where a negative relationship is expected with these factors. Lastly there is school staffing where student teacher ratios may differ based on school district resources and funding. With this the expected relationship would be positive as students do better the more attention they can get from an educator.

Methodology

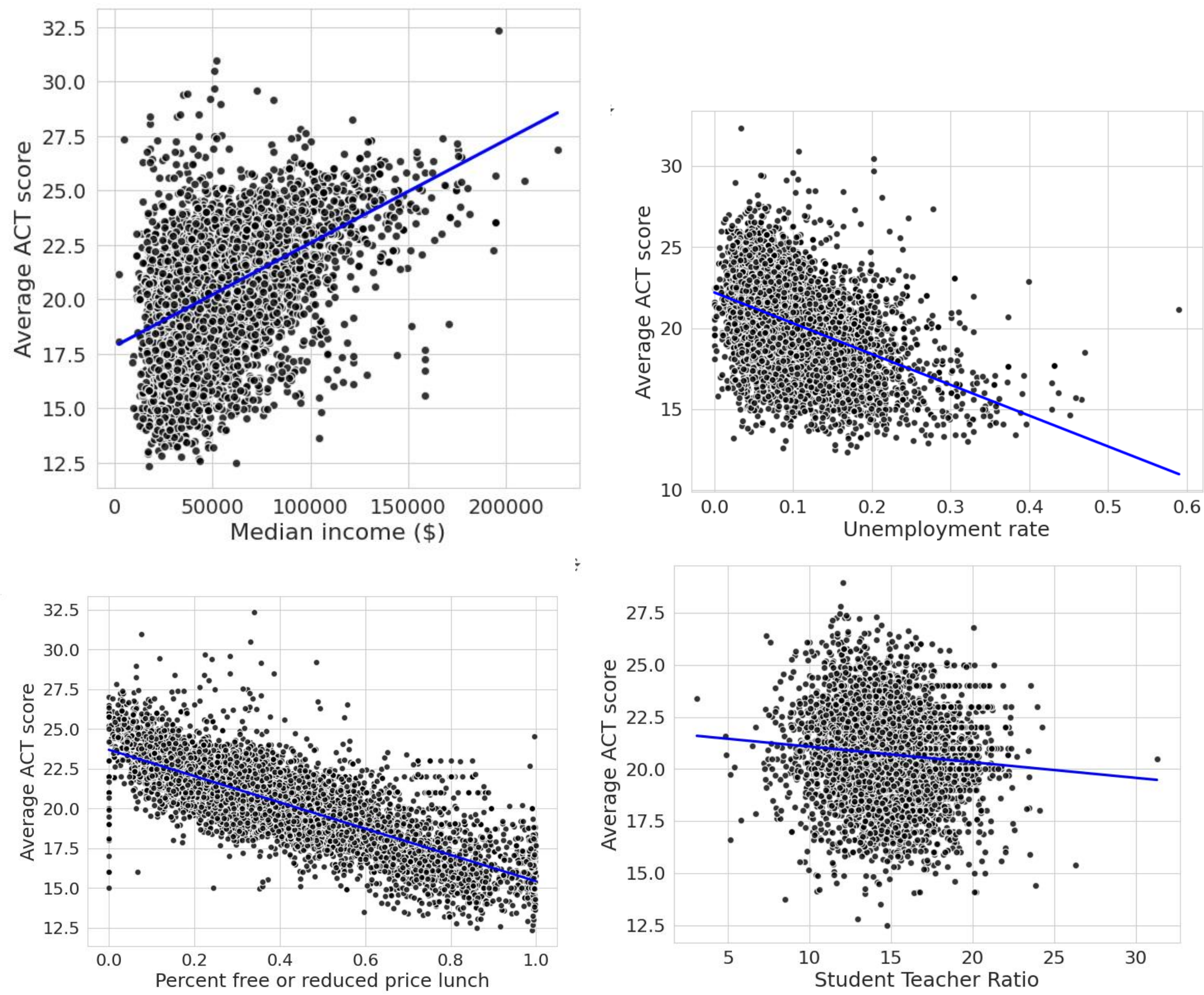
To explore relationships, I first generated correlation matrices and scatterplots. I then fit ordinary least squares linear regression models for each predictor individually, followed by a multiple regression model including all socioeconomic variables. Because linearity appeared appropriate and quadratic terms were not significant in most cases, the final models used simple linear terms. I evaluated model fit using r-squared, p-values, residual plots, and mean absolute error to compare prediction accuracy. An ANOVA model comparison was used to test whether removing weak predictors reduced explanatory power. Finally, I standardized predictors using z-scores with a mean of 0 and standard deviation of 1 to compare effect sizes on a common scale. All analyses were performed in Python using pandas, seaborn, statsmodels, and scikit-learn

Computational Results

A correlation heatmap shows a strong negative correlation between percent free/reduced-price lunch (r = –0.78) and average ACT score. Median income and percent college-educated adults showed moderate positive correlations with ACT scores (r is about 0.44–0.47), while student–teacher ratio and unemployment rate were weakly correlated (|r| < 0.10).



Simple linear regression models were fit for each predictor. The percent lunch model produced the highest r-squared (0.606), followed by median income and percent college (r-squared of about 0.19–0.22). Student–teacher ratio had the lowest correlation (r-squared = 0.01). Residual plots for all single-predictor models did not show lack of linearity. This was shown with quadratic models making a marginal difference at best.



A multiple regression using all socioeconomic variables achieved and r-squared = 0.621. After removing non-significant predictors, the reduced model with percent lunch, unemployment rate, and percent college retained nearly identical explanatory power (r-squared = 0.621) and a similar mean absolute error of about 1.14. ANOVA confirmed that excluding weak predictors did not significantly reduce model fit with p = 0.60. Standardized regression coefficients showed that percent reduced lunch had the largest magnitude effect after normalization. Model error, assessed using mean absolute error, was similar across the full, reduced, and normalized models.

Discussion

The results strongly support the hypothesis that socioeconomic factors are meaningful predictors of school performance. Among all variables tested, the percentage of students receiving free or reduced-price lunch was the strongest predictor of ACT scores. In this dataset, percent reduced lunch alone explained over 60% of the variation in ACT performance, and it remained the dominant term in both the reduced and normalized regression models. This suggests that economic disadvantage is closely tied to lower test scores, potentially through mechanisms such as reduced access to learning resources, higher stress, and decreased academic support outside of school. Median income and the percentage of college-educated adults also showed meaningful positive relationships with ACT scores. These findings are consistent with the idea that community-level educational attainment and financial stability create environments that support academic achievement. The lack of a stronger correlation however is likely due to this being census data and not coming directly from attendees of the school, just surrounding area of the schools. Due to zoning and students not necessarily going to their neighborhood school, these factors may not be as relevant in every case. In contrast, unemployment rate, and student–teacher ratio were much weaker predictors. While unemployment rate remained statistically significant in the reduced model, its effect size was small compared to other variables. The lack of correlation between student–teacher ratio and ACT scores was somewhat unexpected but can be reasonably explained. First, class size effects tend to be subtle and more influential in early grades, while ACT performance reflects cumulative learning through high school. Second, student–teacher ratio is a coarse metric and it does not capture instructional quality, teacher experience, curriculum rigor, or how teachers are distributed across subjects. Third, this dataset averages size across grades, with averages using coming from k-12 data. From this it is hard to extract meaningful relationships especially in relation to high school. As a result, even if the student–teacher ratio has a real educational impact, it may not be detectable in this model. Overall, the regression results demonstrate that poverty-linked variables are the primary drivers of school performance in this dataset. The strong agreement among the full, reduced, and normalized models increases confidence in the findings. However, this analysis is observational, and causation cannot be assumed. Additionally, the dataset does not include potentially important variables such as school funding, teacher quality, or student demographics.

Conclusion

The purpose of this study was to determine whether school performance, measured by average ACT score, could be explained by socioeconomic conditions in U.S. school districts. Despite limitations in the data, the evidence clearly supports the conclusion that socioeconomic disadvantage, most directly reflected through free and reduced-price lunch rates, is the most powerful predictor of ACT performance in this sample. Policies aimed at reducing economic barriers for students may therefore have the greatest potential to improve academic outcomes.

References

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